



Alberta Public Health Disease Management Guidelines

Lymphogranuloma Venereum



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Health and Wellness Promotion Branch

Public Health and Compliance Branch

Alberta Health

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Case Definition

Confirmed Case

Laboratory confirmation with or without clinical illness^(A) in appropriate specimen (e.g., oropharyngeal, urethral, rectal or lesion swab; bubo aspirate; urine or blood sample):

- DNA sequencing or RFLP (Restriction fragment length polymorphism) of *Chlamydia trachomatis* confirming serovars of L1, L2, or L3

Probable Case

Isolation or detection of *C. trachomatis* by culture, antigen or serology (MIF $\geq$ 1:256, CF $\geq$ 1:64)

AND one of the following:

- Proctitis

OR

- Inguinal/femoral lymphadenopathy

OR

- Epidemiological link to a confirmed or probable case of Lymphogranuloma Venereum (LGV).

Note: Cases which fit a probable case definition but test negative for LGV serovars on confirmatory (genotype) testing are not considered probable cases; cases which fit a probable case definition and whose test results are inconclusive on confirmatory (genotype) testing are considered probable cases.

Note: Where possible, suspected cases of LGV should have both a swab and sera submitted for laboratory testing. Please contact your local laboratory or the National Microbiology Laboratory for more information and advice regarding specimen collection and transport.

***NAAT is not officially approved in Canada for use with rectal or oropharyngeal swabs.** However, some local laboratories in Alberta have conducted internal validations and routinely offer NAAT for Chlamydia testing from rectal specimens.

^(A) Clinical illness is characterized by proctitis, inguinal and femoral lymphadenopathy $\pm$ genital ulcer.

Reporting Requirements

Note: This section includes First Nations and Inuit Health Branch

Physicians, health practitioners and others shall notify the Sexually Transmitted Infection (STI) Medical Director^(B) via Sexually Transmitted Infection Centralized Services (STICS), of all confirmed and probable cases within 48 hours (two business days) by forwarding a completed [Notification of STI](#) form.

Laboratories

All laboratories shall report all positive laboratory results by mail, fax or electronic transfer within 48 hours (two business days) to the:

- STI Medical Director via STICS, and
- Chief Medical Officer of Health (CMOH) (or designate).

Alberta Health Services - STICS

Contact Information: Toll free: 1-855-945-6700 option 4
Fax: 780-670-3624

The STI Medical Director/STICS shall forward the initial Notification of STI form of all confirmed cases to the CMOH (or designate) within two weeks of notification and the final Notification of STI form within four weeks. The STI Medical Director/STICS are responsible for ensuring investigation and follow-up of all reported confirmed cases.

- For out-of-province and out-of-country reports, the following information (when available) should be forwarded to the CMOH (or designate) by phone, fax or secure/encrypted electronic transfer as soon as possible:
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - positive laboratory report, and
 - other relevant clinical/epidemiological information.
- For out-of-province and out-of-country contacts the following information (when available) should be forwarded to the CMOH (or designate) as soon as possible:
 - name,
 - date of birth,
 - date of exposure, and
 - out-of-province/country contact information.

^(B) The STI Medical Director is the Provincial Medical Director of Alberta Health Services' Sexually Transmitted Infection Centralized Services (STICS) and is also a Medical Officer of Health.

Additional Reporting Requirements for Health Practitioners

- An [LGV Enhanced Surveillance Form](#) should be completed and submitted to the CMOH for all probable and confirmed cases of LGV. Refer to the [Public Agency of Canada website](#) for more information.
- In all cases, where a person under 18 is suspected or confirmed to have a STI, an assessment should be carried out by the clinician to determine if additional reporting is required.

To Alberta Child and Family Services

- The clinician should determine whether there are reasonable and probable grounds to believe that they are in contact with “a child in need of intervention” (as per Section 1(2) of the [Child, Youth and Family Enhancement \(CYFE\) Act](#)) and shall report to a director pursuant to Section 4 of the *CYFE Act*.⁽¹⁾
- Reporting is done by contacting the local Child and Family Services office or calling the **CHILD ABUSE HOTLINE: 1-800-387-5437 (KIDS)**. For local office contact information see [the ministry's website](#).

To Law Enforcement Agency

- Consent is a key factor in determining whether any form of sexual activity is a criminal offence. Children under 12 do not have the legal capacity to consent to any form of sexual activity. The law recognizes that the age of consent for sexual activity is 16. However, the law identifies the exception for minors between 12 and 16 years as having the ability to consent, in “close in age” or “peer group” situations.⁽²⁾
- Reporting is done by contacting your local City Police Detachment or [RCMP Detachment](#).
- For additional information:
 - Alberta [Child, Youth and Family Enhancement Act](#)
 - Age of Consent to Sexual Activity at www.justice.gc.ca/eng/rp-pr/other-autre/clp/faq.html
 - Criminal Code of Canada at www.laws-lois.justice.gc.ca/eng/acts/C-46/

Epidemiology

Etiology

Lymphogranuloma venereum is caused by *Chlamydia trachomatis*; a bacterial agent with approximately 18 serologic variants (serovars) divided between two biologic variants (biovars): oculogenital (serovars A – K) and LGV (serovars L₁, L₂, and L₃). Lymphogranuloma venereum is caused by the biovar LGV, serovars L₁, L₂, and L₃. These strains invade and reproduce in regional lymph nodes.^(3,4)

Clinical Presentation

There are three distinct stages of infection with LGV; primary, secondary and tertiary. The disease begins with a small painless papule, nodule or lesion at the site of inoculation (penis, vulva, vagina, rectum, oral cavity or occasionally cervix) that heals quickly and may go unnoticed (asymptomatic) in up to 50% of people. This appears 3–30 days after contact.⁽³⁾

The second stage occurs within two to six weeks of the primary lesion, and includes the development of lymphadenopathy and/or anus and rectum involvement. The first obvious symptom of infection is tender enlargement of inguinal/femoral lymph nodes. Cervical lymph nodes may also be inoculated after oral sex. This lymphadenopathy is accompanied by significant systemic symptoms (low grade fever, chills, myalgias, arthralgias, etc.). Abscesses and draining sinuses may also occur (less than 33% of cases). Involvement of the anorectum results in bloody, purulent or mucous discharge from the anus as well as constipation.⁽³⁾

In the third stage, chronic inflammation leads to the destruction of tissue in the involved area. This inflammation contributes to lymphatic obstruction which results in genital elephantiasis, genital and rectal strictures and fistulae. This occurs in approximately 10–20% of untreated cases, and occurs more commonly in females.⁽³⁾

Diagnosis

- Diagnosis is not always straightforward as the symptoms and signs of LGV may overlap with other STI. The diagnosis is based on history and clinical features with confirmatory laboratory testing.

Treatment

Preferred

- Doxycycline 100mg orally twice a day x 21 days (B-II)

Alternate

- Erythromycin 500 mg orally four times a day x 21 days (C_III) (erythromycin dosage refers to use of erythromycin base), or
- Azithromycin 1 gm orally once a week for three weeks (C_III)
 - Note: While some experts believe Azithromycin to be effective in the treatment of LGV, clinical data is lacking.⁽³⁾

Pediatric Cases

- Neonates born to untreated, infected mothers must be tested and treated.
- If the case is in an infant, the mother and her sexual partner(s) should be examined and tested.
- It is recommended that all children < 14 years of age be referred to a pediatrician and, because of the high risk of sexual abuse (excepting those < one month of age with a conjunctivitis or < six months of age with pneumonia), be managed in consultation with one of the following referral centres.

Edmonton

Child and Adolescent Protection Centre
Stollery Children's Hospital
1C4.24 Mackenzie Health Sciences Centre
8440-112 Street
Edmonton, Alberta T6G 2B7
Tel: 780-407-1240

Calgary

Child Abuse Service
Child Development Centre
Suite 200, 3820-24 Ave NW
Calgary, Alberta. T2N 1N4
Tel: 403-955-5959

Reservoir

The only known reservoir is humans.⁽⁵⁾

Transmission

Direct vaginal, anal or oral sexual contact with open lesions of infected persons.⁽⁵⁾

Incubation Period

The incubation period is variable ranging from three to 30 days for a primary lesion. If buboes are the first identified symptom, the incubation period is two to six weeks.⁽³⁾

Period of Communicability

The infection is communicable during the presence of active lesions. This is variable, lasting weeks to years.⁽⁵⁾

Host Susceptibility

Susceptibility is universal. It is uncertain if an individual may develop natural or acquired immunity.⁽⁵⁾

Incidence

General

Until relatively recently, LGV was a rare infection in industrialized countries and was usually acquired in endemic areas. LGV is endemic to parts of Africa, Asia, South America and the Caribbean – where it accounts for an estimated 2–10% of genital ulcer disease. Since 2003, cases have been reported among men who sleep with men (MSM) in Belgium, France, Germany, Sweden, the United Kingdom, the United States and Canada.^(3,4)

Canada

LGV is not nationally reportable. However, an enhanced surveillance system for LGV was initiated by the PHAC, in partnership with the provincial and territorial governments in February 2005.⁽³⁾

Between November, 2001 and December, 2009 there have been 112 cases of LGV reported in Canada; 65 are confirmed cases and 47 are probable cases (according to the national case definition). All reported cases to date have been male, and most cases report recent, unprotected sexual contact with male partners primarily in bathhouses.⁽⁶⁾

Alberta

No cases were reported in the period from 1998 to 2004. In 2005 and 2006, one case of LGV was reported each year. Both cases were MSM and one was HIV co-infected. No further cases have been identified since 2006.⁽⁷⁾

Refer to the [Interactive Health Data Application](#) for more information.

Public Health Management

Key Investigation

The diagnosis and treatment is performed by community health care providers.

- Determine the presence or absence of symptoms.
- Determine if risk factors for LGV are present:
 - sexual contact with person(s) with known infection or compatible syndrome,
 - unprotected sex with partner(s) originating from or following travel to an area with high endemicity for LGV,
 - travelers to endemic countries who have had unprotected sex with a resident of that area, and
 - men who have unprotected sex with other men.
- Offer testing for HIV and other STI.
- Counsel and identify partners, including locating information.

Management of a Case

- Cases should be interviewed for history of exposure, risk assessment and sexual partner(s) identification.
- Testing for chancroid and granuloma inguinale should also be considered in individuals with lesions who have traveled to, or have a sexual partner(s) from, areas endemic for these infections.
- All cases should be instructed about infection transmission. Patients should be counseled about the importance of abstaining from unprotected intercourse until seven days after completion of treatment by both case and partner(s).
- All cases should be provided with individualized STI prevention education, targeted at developing knowledge, skills, attitudes and behaviors to reduce the risk and prevent recurrences of STI.
- Immunization against hepatitis A may be recommended. Refer to Alberta Immunization Policy for immunization eligibility.
- Immunization against hepatitis B is recommended if not already given. Refer to Alberta Immunization Policy for immunization eligibility.
- All patients with a notifiable STI qualify for provincially funded medications.
 - STICS will send replacement medication upon receipt of a Notification of STI Form when the health care provider mailing address is indicated on the form.
 - Health care providers may order additional quantities of most medications by contacting STICS.
- For advice on management of your case, call **STICS toll free 1-855-945-6700, option 4.**
- Sexual assault in adults should be managed in conjunction with local Sexual Assault services and other appropriate community support services.

Recalcitrant Patients

The [Public Health Act](#) (sections 39 through 52) authorizes detention of recalcitrant patients for medical examination, treatment and/or counselling.

- The CMOH (or designate) or MOH (or designate) may issue a certificate to detain an individual who is believed to be infected and refuses or neglects to comply with treatment.
- In order to enact this provision, there must be proof of infection or contact with an infected person and:
 - documentation of failure to comply with prescribed treatment and medical examination, or
 - non-compliance for testing and/or treatment.

Management of Contacts

Partner Notification

- Partner notification will identify those at risk, reduce disease transmission/re-infection and ultimately prevent disease sequelae.
- **It is mandated under the *Communicable Diseases Regulation* that every attempt is made to identify, locate, examine and treat partners/contacts of all cases.**
- All sexual contacts in the last 60 days, regardless of signs or symptoms, must be located, examined, tested and treated empirically as follows:
 - Azithromycin 1 gm po in a single dose or doxycycline 100 mg po bid for seven days.
 - If tests confirm an LGV infection, re-treat as recommended for cases (see [Treatment](#)).
- Contacts should abstain from unprotected intercourse with the case until the treatment of the case is complete.
- All contacts should be:
 - screened for HIV and other STI,
 - instructed about infection transmission, and
 - provided with individualized STI prevention education targeted at developing knowledge, skills, attitudes and behaviors to reduce the risk and prevent recurrences of STI.
- Health care providers are required to provide partner names and locating information on the Notification of STI form and forward to STICS.
- If testing and/or treatment of partner(s) are not confirmed on the Notification of STI form, STICS will initiate follow up by a Partner Notification Nurse (PNN).
 - PNNs are specially trained to conduct notification of partners and contacts in a confidential manner that protects the identity of the index case.
 - The phone number for your designated PNN is available by calling **STICS toll free 1-855-945-6700, option 4.**
 - STICS will follow-up on any incoming referrals of cases and partner(s) from all out of province/country referrals.

Preventive Measures

- Ensure appropriate treatment of LGV for cases.
- Interview case and identify and ensure appropriate treatment of LGV for sexual partner(s).
- Ensure STI care is culturally appropriate, inclusive, readily accessible, and acceptable.
- Include information about risk for STI during pre-travel health counseling.
- Educate the case, sexual partner(s) and the public about symptoms, transmission and prevention of infection including:
 - personal protective measures including the correct and consistent use of condoms
 - abstinence,
 - delaying onset of sexual activity,
 - developing mutually monogamous relationships,
 - reducing the numbers of sexual partners,
 - discouraging anonymous or casual sexual activity, and
 - sound decision making.

Appendix 1: Revision History

Revision Date	Document Section	Description of Revision
October 2021	General	- Updated Template - Etiology, Clinical Presentation, Diagnosis and Treatment sections moved to Epidemiology - Key Investigation section moved to Public Health Management (formerly called Control) - Updated web links and added links where applicable
	Reporting	Revised to reflect current practice
	References	Deleted references that are now hyperlinks and renumbered accordingly

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